

MANUEL RODRÍGUEZ VILLEGAS

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PROFILE

Artificial Intelligence Master's student with a BE in Mathematical Engineering and Artificial Intelligence.
International experience (research and industry) in Machine Learning, Computer Vision, and Deep Learning.

EDUCATION

Comillas Pontifical University, ETSI ICAI <i>Master's Degree in Artificial Intelligence; GPA: 9.08/10; Class Representative</i>	Madrid, Spain 2025 – 2026
Cornell University <i>Exchange Student - Electrical & Computer Engineering</i>	Ithaca, NY, USA Jan – May 2025
Comillas Pontifical University, ETSI ICAI <i>BE in Mathematical Engineering and AI; GPA: 8.75/10</i>	Madrid, Spain 2021 – 2025

EXPERIENCE

AUDI AG <i>Geometric AI Intern</i> Working on 3D structures classification and process development optimization with agentic AI.	Ingolstadt, Germany July 2026 – Nov 2026
Comillas Pontifical University <i>Teaching Assistant</i> Contributed to deployment of a GPU-based AI cluster to support student workloads in deep learning courses.	Madrid, Spain Sep 2025 – Apr 2026
Azzulei Technologies <i>Computer Vision Intern</i> Built a real-time multi-object tracking pipeline (YOLOv8 + Optical Flow) for automated sports camera control, achieving 30 FPS on GPU and reducing manual intervention by 70%.	Madrid, Spain Jun – Aug 2025
Imperial College London <i>Research Assistant</i> Reproduced nonlinear dynamical systems using Neural ODEs and PINNs (PyTorch), validating stability and trajectory accuracy against ground-truth simulations and different initial conditions.	Remote (London, UK) Jun – Aug 2024
Endesa <i>Energy Data Analyst Intern</i> Forecasted daily Iberian ancillary service costs using MLP and regression models (MAPE < 10%).	Madrid, Spain Jun – Aug 2023

BACHELOR THESIS

Visual Topological SLAM using Deep Learning Techniques - Developed a visual topological SLAM system in ROS2 using CNN-based place recognition. - Integrated voice-controlled navigation interface with NLP and achieved robust localization without metric maps.	2025
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SKILLS

Programming: Python (advanced), R, C#, SQL, C++ (basic)
Tools: PyTorch, Git, Docker, CI/CD, HuggingFace, ROS 2, Kubernetes (K8s), OpenCV, MATLAB, n8n
AI & ML: Deep Learning, Generative AI, Computer Vision, NLP, MLOps, Deep Reinforcement Learning
Mathematics: Linear Algebra, Analysis, Optimization, Dynamical Systems, Statistics

LANGUAGES

Spanish (Native) | English (Fluent) | German (Beginner)

HONORS & AWARDS

Hackathon Winner <i>Developed the winning project in the 10th Smart Industry Hackathon, featuring Kearney and CAF.</i>	2025
UNIJES Social Tech Challenge Winner <i>Voice-controlled wheelchair project</i>	2024
Academic Excellence Scholarship <i>Top-scoring students across Community of Madrid</i>	2021, 2023, 2024
Baccalaureate Honorific Mention <i>2nd highest GPA in Community of Madrid</i>	2021